Chase Blagden

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CBlagden
US Citizen

Education

2019 – 2023 **California Institute of Technology**, B.S. in Computer Science.

- **Courses:** • Introduction to Data Structures and Algorithms • Introduction to Embedded Systems • Introduction to Computer Programming • Mechanical Prototyping
- **Involved in:** Robotics Team - Using OrbSLAM with ROS to accurately map an underwater environment.

2015 – 2019 **Servite High School**.

- **Awards:** • St. Alexis Award for Outstanding Scholarship • Bausch and Lomb Honorary Science Award

Experience

Winter 2020 - Present **Caltech**, *Center for Autonomous Systems and Technology*.

- Developing fully autonomous vehicles by encompassing vehicle dynamics and perception with a full sensor suite using Lidar, camera and thermal imaging for object detection and semantic segmentation imaging.

Winter 2020 **Caltech**.

- Developed model finetuned with past articles on GPT-2 to produce articles for student newspaper reaching 2000 students

Fall 2019 **Undergraduate Research**, *Caltech*.

- Developed model to predict reagents of chemical reactions using gradient-boosted decision trees using AWS.

Summer 2019 **Freshman Summer Research Institute**, *Caltech*.

- Studied the effects of elasticity and near-field compressibility on bubble oscillations via implementation of a system of nonlinear, coupled differential equations in MATLAB.

2016 – 2019 **VP of Software Development**, *FRC Team 3309*.

- Won competition of 30+ competing teams and ranked 5th place at the world competition.
- Implemented 1st place ranking autonomous visual servoing system to deliver game objects to target

2018 – 2019 **VP of Human Resources**, *Trinity Corp*.

- Oversaw 400+ students to complete mandated work-study hours for financial aid with $1 million dollars in revenue

Summer 2018 **Beaver Works Summer Institute**, *Student*, MIT.

- Developed autonomous remote-control car operating on Jetson TX2 with ROS using OpenCV and a potential field controller using LiDAR to complete obstacle-laced course.
- Led team to win 2nd Place in Grand Prix out of 14 teams.

Skills

Programming Languages: Python, Java, MATLAB, C++

Frameworks: Pytorch, Keras, NumPy, Git, OpenCV, Robot Operating System